	GLOBAL ECONOMICS **Grade 10** **Catch ups exam** **Decision-making under uncertainty** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective:
Characterise economic decisions under uncertainty, organise payoff tables, calculate profits from revenues and costs, and justify choices using Maximax, Maximin, Minimax Regret, EMV/Bayes, and probability sensitivity analysis.

Assessment criteria

C1. Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C2. Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.

C3. Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison.

C4. Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

C5. Constructs opportunity loss tables and applies the Minimax Regret criterion to select the alternative that minimises the maximum opportunity loss.

C6. Calculates Expected Monetary Value using stated or estimated probabilities and applies the Bayes decision criterion to select the alternative with the highest expected payoff.

C7. Uses subjective probabilities and sensitivity analysis to examine how changes in probability assumptions affect the preferred decision.

Criteria assessment

This exam assesses C1–C7 for a total of **52 points**. Task labels show the points available. A minimum of **4 points in each criterion** is required to pass that criterion.

Show the required procedure and all calculations. Answers without the required procedure receive no credit. Partial credit may be awarded for a correct method with arithmetic errors. Answer on separate paper and number your answers.

Question	C1	C2	C3	C4	C5	C6	C7
1	1	2	–	–	–	–	–
2	1	3	–	–	–	–	–
3	2	5	–	–	–	–	–
4	1	–	2	–	–	–	–
5	1	–	3	–	–	–	–
6	2	–	5	–	–	–	–
7	–	–	–	2	–	–	–
8	–	–	–	2	–	–	–
9	–	–	–	2	–	–	–
10	–	–	–	–	2	–	–
11	–	–	–	–	2	–	–
12	–	–	–	–	2	–	–
13	–	–	–	–	–	2	–
14	–	–	–	–	–	2	–
15	–	–	–	–	–	2	–
16	–	–	–	–	–	–	2
17	–	–	–	–	–	–	2
18	–	–	–	–	–	–	2
Total	**8**	**10**	**10**	**6**	**6**	**6**	**6**

1. Book Fair Stall [C1-1 C2-2]

A bookseller must choose a small stall or a large stall before attendance at a weekend fair is known. Attendance will be either high or low. A small stall earns a profit of 18 thousand USD with high attendance and 11 thousand USD with low attendance. A large stall earns 34 thousand USD with high attendance and 3 thousand USD with low attendance.

(C1) Characterise the decision: identify the alternatives, states of nature, consequences, and economic context. **[1]**

(C2) Construct the payoff table with labelled rows, columns, and units. **[2]**

2. Laundry Service [C1-1 C2-3]

A laundry business must choose a self-service plan or a staffed plan before weekly demand is known. Demand will be high or low. The self-service plan earns a net contribution of 3 USD per load and handles 160 loads with high demand or 100 with low demand. The staffed plan earns a net contribution of 5 USD per load and handles 130 loads with high demand or 70 with low demand. For either plan, high demand brings a fixed weekly bonus of 40 USD, while low demand causes a fixed weekly loss of 30 USD. These fixed amounts are additional to the per-load contributions; there are no other adjustments.

(C1) Identify the alternatives, states, consequences, and economic context. **[1]**

(C2) Calculate the payoffs and construct a labelled payoff table with units. **[3]**

3. Craft Workshop [C1-2 C2-5]

A craft studio must choose a pottery workshop or a weaving workshop before tourist demand is known. Demand will be strong or weak. Pottery provides a net contribution of 12 USD per participant and a fixed sponsorship contribution of 80 USD per workshop. Weaving provides 15 USD per participant and a fixed sponsorship contribution of 50 USD per workshop. Strong demand adds a further net contribution of 6 USD per participant to either workshop; weak demand adds 2 USD per participant. Pottery attracts 30 participants with strong demand or 18 with weak demand; weaving attracts 24 or 16, respectively. All contributions are additive, and no other adjustments are needed.

(C1) Identify the alternatives, states, consequences, and economic context; explain which elements the studio controls and which it does not. **[2]**

(C2) Calculate all four payoffs, showing how you combine the contributions, and construct a labelled payoff table with units. **[5]**

4. Cinema Programme [C1-1 C3-2]

A cinema must choose a film marathon or a family screening before weekend attendance is known. Attendance will be high or low. All amounts are in thousand USD. For the marathon, revenue is 64 and total cost is 26 with high attendance; revenue is 30 and total cost is 24 with low attendance. For the family screening, revenue is 50 and total cost is 23 with high attendance; revenue is 36 and total cost is 21 with low attendance. Each total cost includes all costs for that outcome.

(C1) Identify the alternatives, states, consequences, and economic context. **[1]**

(C3) Calculate every profit and construct a labelled payoff table with units. **[2]**

5. School-Bag Production [C1-1 C3-3]

A workshop must choose canvas bags or recycled-fabric bags before market conditions are known. The two possible conditions are strong demand with cheap materials, or weak demand with expensive materials. Canvas-bag sales are 80 bags in the first state and 50 in the second; recycled-fabric bag sales are 70 and 40, respectively. Revenue per bag is 24 USD for canvas and 28 USD for recycled fabric. Variable cost per bag is 10 USD in the first state and 16 USD in the second, for either product. All bags produced are sold, and there are no fixed or other costs.

(C1) Identify the alternatives, states, consequences, and economic context. **[1]**

(C3) Calculate revenues, costs, and profits for every outcome; construct a labelled profit payoff table with units. **[3]**

6. Guided Cycling Tours [C1-2 C3-5]

A tour operator must choose a town route or a countryside route before seasonal demand is known. Demand will be strong or weak. The town route sells 90 places with strong demand or 60 with weak demand; the countryside route sells 70 or 40, respectively. Base revenue per place is 32 USD for the town route and 45 USD for the countryside route. Strong demand adds a surcharge of 8 USD per place to either route; weak demand adds 3 USD per place. Variable cost per place is 18 USD for the town route and 26 USD for the countryside route. Total fixed seasonal costs, for either route, are 500 USD with strong demand and 650 USD with weak demand. These fixed costs are paid once per season, and there are no other revenues or costs.

(C1) Identify the alternatives, states, consequences, and economic context; distinguish the controlled choice from the uncertain demand. **[2]**

(C3) Calculate total revenues, total costs, and profits for all four outcomes. Construct a labelled profit payoff table with units. **[5]**

7. Sportswear Stock [C4-2]

A retailer must choose a seasonal stock plan before demand is known. Demand will be high, medium, or low. Profits are in thousand USD; no probabilities are available.

Alternative	High	Medium	Low
Bulk order	120	48	-30
Regular order	84	57	16
Small order	55	43	29
Pre-orders only	24	24	24

(C4) Calculate each alternative's maximum and minimum payoff. Select the Maximax and Maximin decisions, and briefly interpret the risk trade-off. **[2]**

8. Concert Venue [C4-2]

An organiser must book one venue before ticket demand is known. Demand will be high, medium, or low. Profits are in thousand USD; no probabilities are available.

Alternative	High	Medium	Low
Stadium	150	42	-45
Theatre	95	62	12
Community hall	60	46	26
Online event	30	30	30

(C4) Calculate each alternative's maximum and minimum payoff. Select the Maximax and Maximin decisions, and briefly interpret the risk trade-off. **[2]**

9. Cold Storage [C4-2]

A food distributor must choose a storage plan before next season's market conditions are known. The four possible states are S1: very strong sales, S2: strong sales, S3: steady sales, and S4: weak sales. Profits are in thousand USD; no probabilities are available.

Alternative	S1	S2	S3	S4
Build large	140	72	14	-40
Build medium	100	76	38	8
Rent small	68	56	42	22
Shared storage	48	44	36	28

(C4) Calculate each alternative's maximum and minimum payoff. Select the Maximax and Maximin decisions, and briefly interpret the risk trade-off. **[2]**

10. Flower Stall [C5-2]

A florist must choose a purchasing plan before holiday demand is known. The three possible demand states are high, medium, and low. Profits are in thousand USD; no probabilities are available. All payoffs are to be maximised.

Alternative	High	Medium	Low
Large purchase	78	32	-12
Medium purchase	60	48	18
Small purchase	40	38	30

(C5) Construct the complete opportunity-loss table using each state's best payoff minus each alternative's payoff. Calculate every alternative's maximum regret and select the Minimax Regret decision. **[2]**

11. Parcel Depot [C5-2]

A parcel company must choose a depot plan before shipment demand is known. Demand will be high, normal, or low. Profits are in thousand USD; no probabilities are available. All payoffs are to be maximised.

Alternative	High	Normal	Low
New depot	112	40	-28
Expand depot	86	64	14
Rent space	62	54	34
Current depot	36	36	36

(C5) Construct the complete opportunity-loss table using each state's best payoff minus each alternative's payoff. Calculate every alternative's maximum regret and select the Minimax Regret decision. **[2]**

12. Bakery Output [C5-2]

A bakery must choose its daily output before demand is known. The four possible demand states are S1: very high, S2: high, S3: moderate, and S4: low. Profits are in USD; no probabilities are available. All payoffs are to be maximised.

Alternative	S1	S2	S3	S4
High output	960	560	180	-240
Flexible output	740	660	460	120
Low output	480	500	440	360

(C5) Construct the complete opportunity-loss table using each state's best payoff minus each alternative's payoff. Calculate every alternative's maximum regret and select the Minimax Regret decision. **[2]**

13. Fruit Packing [C6-2]

A cooperative must choose a packing plan before export demand is known. Demand will be strong, steady, or weak, with the probabilities shown. Profits are in thousand USD.

Alternative	Strong	Steady	Weak
$P(S_j)$	0.40	0.35	0.25
Premium packs	90	44	-12
Mixed packs	68	58	24
Basic packs	46	48	38

(C6) Calculate every alternative's EMV using the stated probabilities. Select and justify the Bayes decision. **[2]**

14. Art Market Space [C6-2]

An art-market organiser must choose a space before visitor attendance is known. Attendance will be high, normal, or low, with the probabilities shown. Profits are in thousand USD.

Alternative	High	Normal	Low
$P(S_j)$	0.30	0.40	0.30
Large tent	104	38	-24
Medium tent	78	56	18
Small tent	56	50	36
Existing rooms	28	28	28

(C6) Calculate every alternative's EMV using the stated probabilities. Select and justify the Bayes decision. **[2]**

15. Repair Shop Staffing [C6-2]

A repair shop must choose a staffing plan before service demand is known. The four possible demand states are S1: surge, S2: strong, S3: steady, and S4: slow, with the probabilities shown. Profits are in thousand USD.

Alternative	S1	S2	S3	S4
$P(S_j)$	0.20	0.30	0.30	0.20
Hire full team	120	72	24	-36
Hire part-time	88	76	52	12
Current team	58	60	54	44

(C6) Calculate every alternative's EMV using the stated probabilities. Select and justify the Bayes decision. **[2]**

16. Beach Equipment [C7-2]

A rental firm must choose a large or a compact equipment package before summer demand is known. Demand will be high (S_1) or low (S_2). Profits are in thousand USD.

Alternative	High (S_1)	Low (S_2)
Large (A_1)	96	8
Compact (A_2)	56	32

Let $P(S_1) = p$ and $P(S_2) = 1 - p$.

(C7) Write both EMVs as functions of p. Determine the feasible domain of p, solve the threshold equation, and give all intervals where each alternative is preferred. State the indifference case and include the domain endpoints. **[2]**

17. Online Course Launch [C7-2]

A training business must choose a full launch or a pilot launch before enrolment is known. Enrolment will be high (S_1), moderate (S_2), or low (S_3). Profits are in thousand USD.

Alternative	S_1	S_2	S_3
Full (A_1)	100	36	-8
Pilot (A_2)	52	48	28

Suppose $P(S_1) = p$ and $P(S_2) = p/2$; determine $P(S_3)$ from the sum-to-one condition.

(C7) Write both EMVs as functions of p. Determine the feasible domain of p, solve the threshold equation, and give all intervals where each alternative is preferred. State the indifference case and include the domain endpoints. **[2]**

18. Printing Contract [C7-2]

A printer must choose a standard or a flexible contract before a client's order schedule is known. The schedule will involve an early order (S_1), a revised order (S_2), or a late order (S_3). Profits are in thousand USD.

Alternative	S_1	S_2	S_3
Standard (A_1)	24	60	84
Flexible (A_2)	78	42	36

Suppose $P(S_1) = 2p$ and $P(S_2) = p$; determine $P(S_3)$ from the sum-to-one condition.

(C7) Write both EMVs as functions of p. Determine the feasible domain of p, solve the threshold equation, and give all intervals where each alternative is preferred. State the indifference case and include the domain endpoints. **[2]**